

Name: G.Sandeep

Ht no:2303A52371

Batch:07

Task Description #1 – Privacy in API Usage

```
import requests

api_key = "YOUR_API_KEY"
city = "Hyderabad"

url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={api_key}"
response = requests.get(url)

print(response.json())
```

```
{'cod': 401, 'message': 'Invalid API key. Please see https://openweathermap.org/faq#error401 for more info.'}
```

```
import os
import requests

api_key = os.getenv("WEATHER_API_KEY")
city = "Hyderabad"

url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={api_key}"
response = requests.get(url)

print(response.json())
```

```
{'cod': 401, 'message': 'Invalid API key. Please see https://openweathermap.org/faq#error401 for more info.'}
```

Task 2 – Privacy & Security in File Handling

```
name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

with open("users.txt", "a") as file:
    file.write(f"{name},{email},{password}\n")
```

```
Enter name: trisha
Enter email: trisha@gmail.com
Enter password: trisha@81
```

```
import hashlib

name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

hashed_password = hashlib.sha256(password.encode()).hexdigest()

with open("users_secure.txt", "a") as file:
    file.write(f"{name},{email},{hashed_password}\n")
```

```
Enter name: trisha
Enter email: trisha@gmail.com
Enter password: tri@81
```

Task 3 – Transparency in Algorithm Design (Armstrong Number)

```
def is_armstrong(number):
    temp = number
    digits = len(str(number))
    total = 0

    while temp > 0:
        digit = temp % 10
        total += digit ** digits
        temp //= 10
```

```
return total == number
```

Task 4 – Transparency in Algorithm Comparison

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr
```

```
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]

    return quick_sort(left) + middle + quick_sort(right)
```

Task 5 – Transparency in AI Recommendations

```
def recommend_products(user_interest):
    products = {
        "laptop": ["Gaming Laptop", "Ultrabook"],
        "phone": ["Android Phone", "iPhone"],
        "books": ["Python Programming Book", "AI Basics Book"]
    }

    recommendations = products.get(user_interest, [])

    for product in recommendations:
        print("Recommended:", product)
        print("Reason: Based on your interest in", user_interest)
```